

Mission Integrity System (MIS) — V1

Scenario: Comms Instability

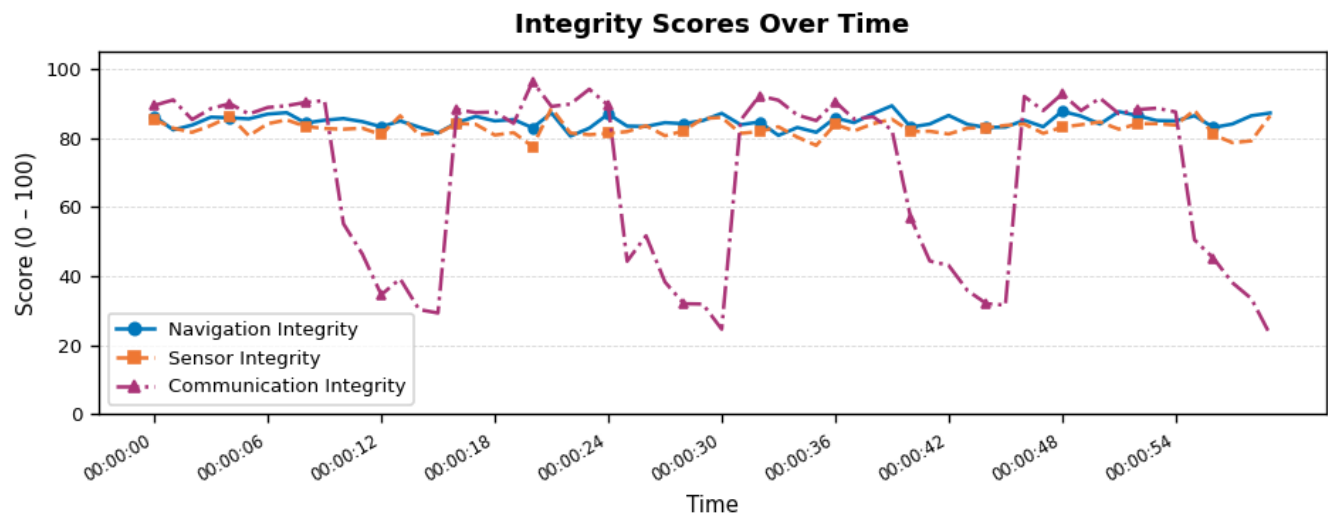
Generated: 2026-04-24 16:49:43 UTC

Demonstration report generated from simulated data

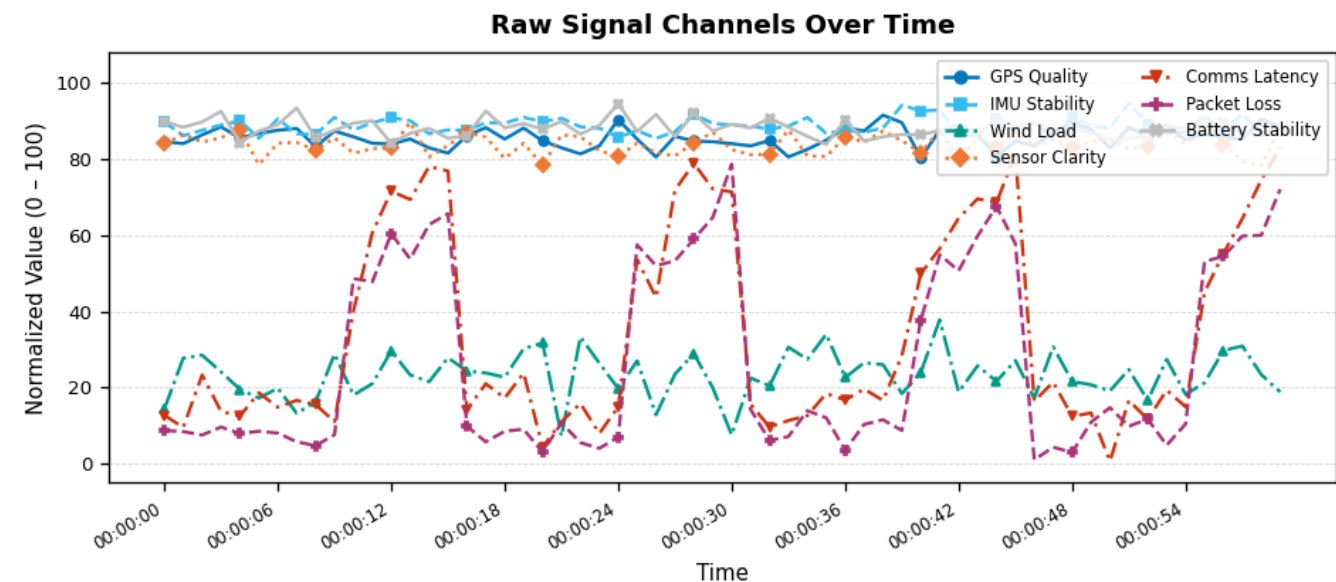
Scenario Summary

Field	Value
Final State	Unreliable
Navigation Integrity	87.3 / 100
Sensor Integrity	86.5 / 100
Communication Integrity	23.0 / 100
Causal Explanation	Unreliable: Communication integrity below minimum trust threshold (< 40).

Integrity Scores Over Time



Raw Signal Channels Over Time



Event Log

Time	Severity	Type	Message
00:00:00	INFO	State Change	Mission state initialized: Stable.
00:00:10	WARNING	State Change	State transition: Stable → Degraded. Degraded: Communication integrity in degraded band (40–59).
00:00:10	WARNING	Threshold Breach	Communication integrity crossed below threshold 75: 90.9 → 55.2.
00:00:10	WARNING	Integrity Drop	Communication integrity dropped 35.7 points in one step: 90.9 → 55.2.
00:00:12	CRITICAL	State Change	State transition: Degraded → Unreliable. Unreliable: Communication integrity below minimum trust threshold (30–39).
00:00:12	CRITICAL	Threshold Breach	Communication integrity crossed below threshold 40: 46.4 → 34.6.
00:00:12	CRITICAL	Integrity Drop	Communication integrity dropped 11.8 points in one step: 46.4 → 34.6.
00:00:16	WARNING	State Change	State transition: Unreliable → Degraded. Degraded: repeated threshold breaches within recent window.
00:00:25	WARNING	Threshold Breach	Communication integrity crossed below threshold 75: 89.5 → 44.3.
00:00:25	WARNING	Integrity Drop	Communication integrity dropped 45.2 points in one step: 89.5 → 44.3.
00:00:27	CRITICAL	State Change	State transition: Degraded → Unreliable. Unreliable: Communication integrity below minimum trust threshold.
00:00:27	CRITICAL	Threshold Breach	Communication integrity crossed below threshold 40: 51.7 → 38.3.
00:00:27	CRITICAL	Integrity Drop	Communication integrity dropped 13.4 points in one step: 51.7 → 38.3.
00:00:31	WARNING	State Change	State transition: Unreliable → Degraded. Degraded: repeated threshold breaches within recent window.
00:00:40	WARNING	Threshold Breach	Communication integrity crossed below threshold 75: 82.4 → 56.8.
00:00:40	WARNING	Integrity Drop	Communication integrity dropped 25.6 points in one step: 82.4 → 56.8.
00:00:41	WARNING	Integrity Drop	Communication integrity dropped 12.4 points in one step: 56.8 → 44.4.
00:00:43	CRITICAL	State Change	State transition: Degraded → Unreliable. Unreliable: Communication integrity below minimum trust threshold.
00:00:43	CRITICAL	Threshold Breach	Communication integrity crossed below threshold 40: 43.1 → 35.8.

Time	Severity	Type	Message
00:00:46	WARNING	State Change	State transition: Unreliable → Degraded. Degraded: repeated threshold breaches within recent window.
00:00:55	WARNING	Threshold Breach	Communication integrity crossed below threshold 75: 87.6 → 50.5.
00:00:55	WARNING	Integrity Drop	Communication integrity dropped 37.1 points in one step: 87.6 → 50.5.
00:00:57	CRITICAL	State Change	State transition: Degraded → Unreliable. Unreliable: Communication integrity below minimum trust threshold.
00:00:57	CRITICAL	Threshold Breach	Communication integrity crossed below threshold 40: 45.2 → 38.1.
00:00:59	CRITICAL	Integrity Drop	Communication integrity dropped 10.5 points in one step: 33.5 → 23.0.

Data Excerpt (Last 15 Steps)

Showing the final 15 time steps of enriched scenario output. Scores are on a 0–100 scale.

Time	State	Nav	Sensor	Comms	GPS Qual	IMU Stab	Wind	Clarity	Latency	Pkt Loss	Batt Stab
00:00:45	Unreliable	83.2	83.7	31.6	84.9	87.1	27.1	87.4	81.4	57.8	88.6
00:00:46	Degraded	85.2	84.1	92.1	83.5	89.6	17.2	83.3	16.4	1.0	89.8
00:00:47	Degraded	83.3	81.4	88.0	87.1	86.2	30.7	85.2	21.4	4.3	87.3
00:00:48	Degraded	87.8	83.2	92.8	89.5	91.3	21.6	83.2	12.4	2.9	88.1
00:00:49	Degraded	86.4	83.9	88.0	87.9	88.7	20.7	84.6	13.3	10.9	87.1
00:00:50	Degraded	84.1	84.7	91.6	82.9	88.2	19.1	85.4	0.6	14.7	85.0
00:00:51	Degraded	87.7	82.6	87.1	88.4	94.7	24.7	82.7	16.6	9.8	85.5
00:00:52	Degraded	86.5	84.2	88.3	86.0	89.3	16.6	83.3	11.6	11.7	86.3
00:00:53	Degraded	85.1	84.2	88.7	89.0	86.8	27.4	88.4	19.3	4.8	88.7
00:00:54	Degraded	85.0	83.8	87.6	85.1	87.1	18.2	83.8	14.9	10.4	88.1
00:00:55	Degraded	86.5	88.1	50.5	86.6	91.4	21.2	91.1	45.0	53.1	87.3
00:00:56	Degraded	83.1	81.0	45.2	85.8	87.0	29.7	83.9	55.2	54.5	89.7
00:00:57	Unreliable	84.1	78.7	38.1	85.3	92.0	30.9	79.4	64.4	59.8	86.8
00:00:58	Unreliable	86.5	79.2	33.5	90.5	86.5	23.4	78.4	74.5	60.0	86.1
00:00:59	Unreliable	87.3	86.5	23.0	88.6	89.2	18.8	88.1	83.1	72.1	88.6

Limitations and Non-Goals

Mission Integrity System (MIS) V1 uses simulated telemetry-like data for demonstration and portfolio purposes only. It is not a UAV control system, flight application, autopilot, navigation engine, targeting system, safety-certified tool, predictive model, or production deployment. MIS V1 performs deterministic classification only and does not recommend or automate operational decisions.